PBHLTH 245 Data Analysis

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High-Level Data Cleaning

```
#remove random/noise variables
healthcare_risk_clean <- healthcare_risk %>%
  select(-c(random_notes, noise_col))

write.csv(healthcare_risk_clean, "healthcare_risk.csv", row.names = FALSE)

#remove any rows with missing entries
healthcare_risk_clean <- healthcare_risk_clean %>% drop_na()

#select only continuous predictors for the first factor analysis
healthcare_risk_cont <- healthcare_risk_clean %>%
  select(Age, Glucose, `Blood Pressure`, BMI, `Oxygen Saturation`,
    Cholesterol, Triglycerides, HbA1c,
    `Physical Activity`, `Diet Score`,
    `Stress Level`, `Sleep Hours`)
```

Data Processing

```
#standardize predictors
stand <- function(v) (v - mean(v)) / sd(v)

Xs <- apply(healthcare_risk_cont, 2, stand)

#sample correlation matrix
cor(Xs)
```

```
##                            Age       Glucose Blood Pressure          BMI
## Age                1.000000000  0.1160346896    0.331515632 -0.004796412
## Glucose            0.116034690  1.0000000000    0.059828033  0.079737765
## Blood Pressure     0.331515632  0.0598280327    1.000000000  0.070001409
## BMI               -0.004796412  0.0797377650    0.070001409  1.000000000
## Oxygen Saturation  0.084655118 -0.0002536509    0.007274556  0.063276492
## Cholesterol        0.184651113  0.0044024736    0.257434537  0.125173328
## Triglycerides      0.112718594  0.0654645591    0.190733923  0.189288642
## HbA1c              0.098635109  0.6263202512    0.045615539  0.121734262
## Physical Activity -0.119583379 -0.2319777357   -0.136781388 -0.352972386
## Diet Score        -0.154783514 -0.3133663315   -0.188622679 -0.295878239
## Stress Level       0.168117674  0.0812881577    0.231607547 -0.011456484
## Sleep Hours       -0.084089688 -0.1254876399   -0.151707490 -0.024749328
##                   Oxygen Saturation  Cholesterol Triglycerides       HbA1c
```

```
## Age                     0.0846551175  0.184651113    0.11271859  0.09863511
## Glucose                -0.0002536509  0.004402474    0.06546456  0.62632025
## Blood Pressure          0.0072745556  0.257434537    0.19073392  0.04561554
## BMI                     0.0632764920  0.125173328    0.18928864  0.12173426
## Oxygen Saturation       1.0000000000 -0.028466499   -0.00338315 -0.00787318
## Cholesterol            -0.0284664990  1.000000000    0.17263564  0.02645367
## Triglycerides          -0.0033831503  0.172635644    1.00000000  0.09858938
## HbA1c                  -0.0078731803  0.026453667    0.09858938  1.00000000
## Physical Activity      -0.0210707396 -0.201993721   -0.26139404 -0.28102124
## Diet Score              0.0249976605 -0.209650411   -0.25949110 -0.35191893
## Stress Level           -0.0958973704  0.164092405    0.10474873  0.07888061
## Sleep Hours             0.1760919458 -0.138996600   -0.12375672 -0.13244786
##                    Physical Activity  Diet Score Stress Level Sleep Hours
## Age                      -0.11958338 -0.15478351   0.16811767 -0.08408969
## Glucose                  -0.23197774 -0.31336633   0.08128816 -0.12548764
## Blood Pressure           -0.13678139 -0.18862268   0.23160755 -0.15170749
## BMI                      -0.35297239 -0.29587824  -0.01145648 -0.02474933
## Oxygen Saturation        -0.02107074  0.02499766  -0.09589737  0.17609195
## Cholesterol              -0.20199372 -0.20965041   0.16409241 -0.13899660
## Triglycerides            -0.26139404 -0.25949110   0.10474873 -0.12375672
## HbA1c                    -0.28102124 -0.35191893   0.07888061 -0.13244786
## Physical Activity         1.00000000  0.47024177  -0.08696709  0.14163103
## Diet Score                0.47024177  1.00000000  -0.14158687  0.19640300
## Stress Level             -0.08696709 -0.14158687   1.00000000 -0.20102508
## Sleep Hours               0.14163103  0.19640300  -0.20102508  1.00000000
```

Running Principal Component Factor Analysis

Eigen Decomposition

```
#Eigen decomposition

fit1 <- eigen(cor(Xs))

#eigen vectors and values
fit1_vectors <- fit1$vectors
fit1_values <- fit1$values

rownames(fit1_vectors) <- colnames(healthcare_risk_cont)
```

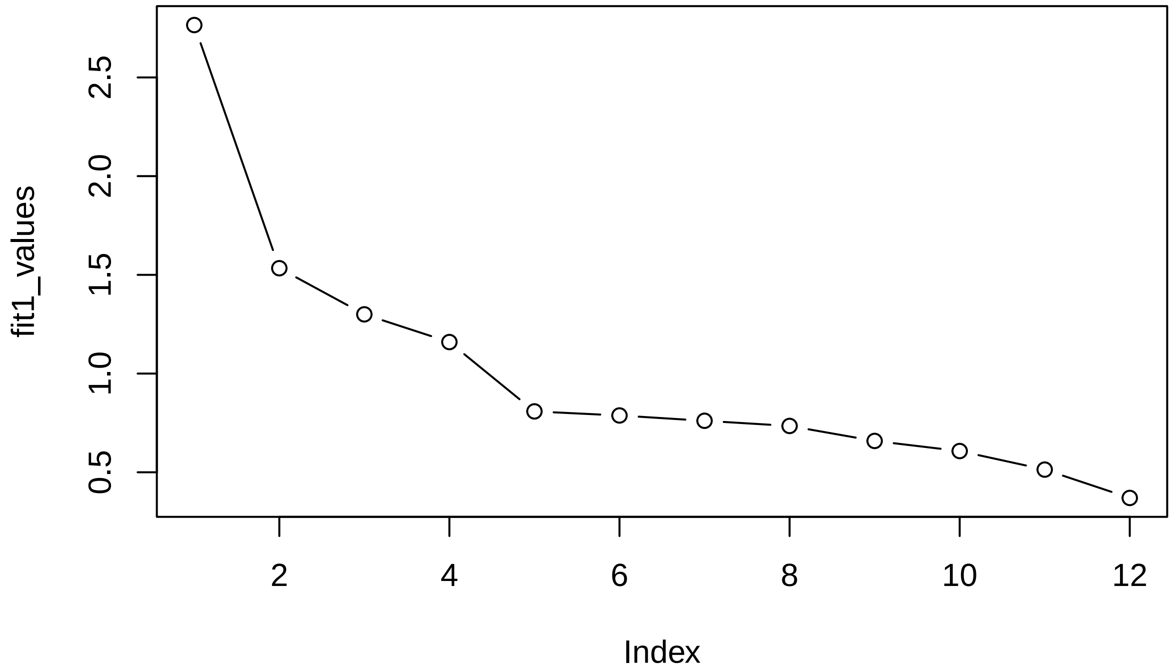
Determing "m"

Before computing loadings for model, wanted to determine appropriate value for m, the number of factors. We want a decent amount of variance to be accounted for with smallest amount of factors.

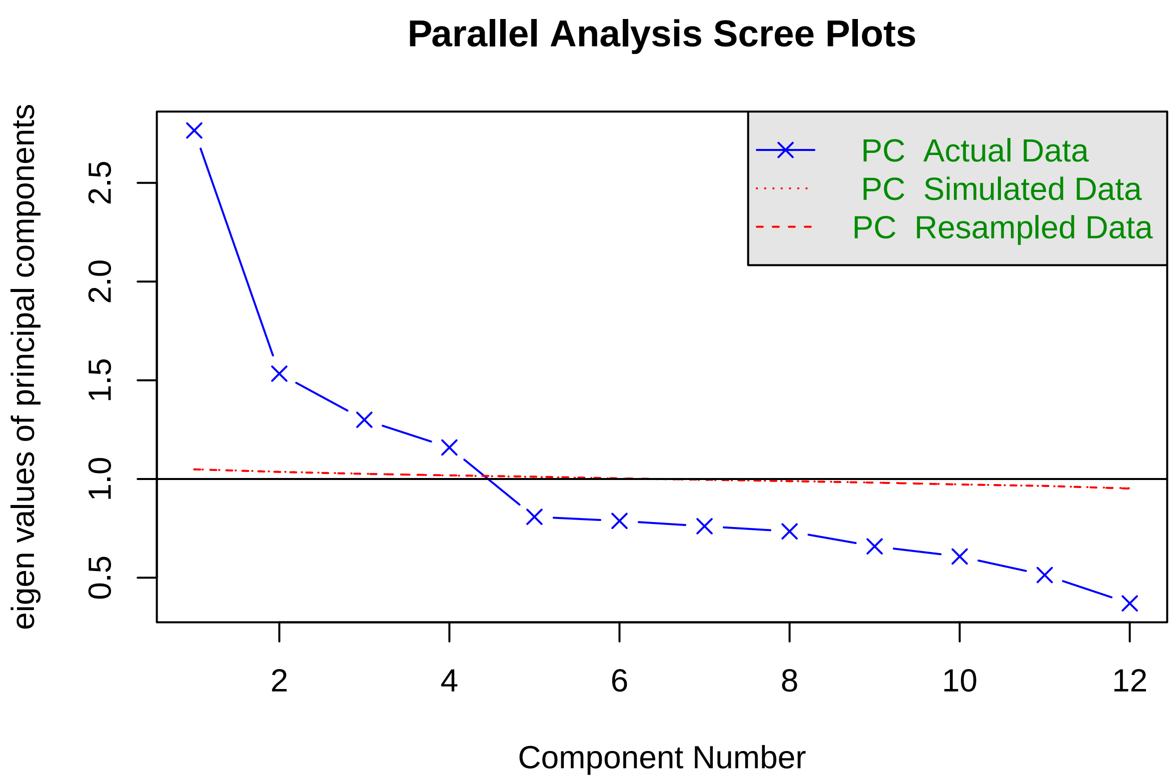
```
library(psych)
```

```
##
## Attaching package: 'psych'

## The following objects are masked from 'package:ggplot2':
##
##     %+%, alpha
```

```r
#scree plot
plot(fit1_values, type = "b")
```




```r
#curve appears to flatten out at m = 5

#parallel analysis (suggested on internet)
fa.parallel(healthcare_risk_cont, fa = "pc")
```



```
## Parallel analysis suggests that the number of factors =  NA  and the number of components =  4
#suggested m = 4
```

Based on these findings, we can run two models (m = 4 and m = 5) and compare them.

Compute Loadings

```r
L1 <- fit1_vectors[,1] * sqrt(fit1_values[1])
round(L1, 3)
```

```
##                Age            Glucose     Blood Pressure                BMI
##             -0.380             -0.543             -0.439             -0.416
## Oxygen Saturation        Cholesterol      Triglycerides              HbA1c
##              0.035             -0.414             -0.462             -0.581
## Physical Activity         Diet Score       Stress Level        Sleep Hours
##              0.664              0.727             -0.345              0.386
```

```r
L2 <- fit1_vectors[,2] * sqrt(fit1_values[2])
round(L2, 3)
```

```
##                Age            Glucose     Blood Pressure                BMI
##             -0.380              0.535             -0.536              0.153
## Oxygen Saturation        Cholesterol      Triglycerides              HbA1c
##              0.105             -0.453             -0.183              0.549
## Physical Activity         Diet Score       Stress Level        Sleep Hours
##             -0.147             -0.139             -0.401              0.199
```

```r
L3 <- fit1_vectors[,3] * sqrt(fit1_values[3])
round(L3, 3)
```

```
##                Age            Glucose     Blood Pressure                BMI
##              0.109              0.373              0.045             -0.583
## Oxygen Saturation        Cholesterol      Triglycerides              HbA1c
##             -0.442             -0.147             -0.289              0.314
## Physical Activity         Diet Score       Stress Level        Sleep Hours
##              0.322              0.122              0.371             -0.389
```

```r
L4 <- fit1_vectors[,4] * sqrt(fit1_values[4])
round(L4, 3)
```

```
##                Age            Glucose     Blood Pressure                BMI
##             -0.555             -0.228             -0.270              0.220
## Oxygen Saturation        Cholesterol      Triglycerides              HbA1c
##             -0.664              0.054              0.163             -0.166
## Physical Activity         Diet Score       Stress Level        Sleep Hours
##             -0.147             -0.111              0.032             -0.382
```

```r
L5 <- fit1_vectors[,5] * sqrt(fit1_values[5])
round(L5, 3)
```

```
##                Age            Glucose     Blood Pressure                BMI
##             -0.009             -0.008              0.031             -0.156
## Oxygen Saturation        Cholesterol      Triglycerides              HbA1c
##              0.111             -0.501              0.703             -0.012
## Physical Activity         Diet Score       Stress Level        Sleep Hours
##              0.087              0.063              0.050             -0.101
```

Compute communalities for m = 4, m = 5 models

```r
comm_m4 <- L1^2 + L2^2 + L3^2 + L4^2
round(comm_m4, 3)
```

```
##                 Age            Glucose     Blood Pressure                BMI
##               0.609              0.772              0.555              0.585
## Oxygen Saturation        Cholesterol      Triglycerides              HbA1c
##               0.648              0.401              0.357              0.765
## Physical Activity         Diet Score       Stress Level        Sleep Hours
##               0.587              0.575              0.419              0.486
```

```r
comm_m5 <- L1^2 + L2^2 + L3^2 + L4^2 + L5^2
round(comm_m5, 3)
```

```
##                 Age            Glucose     Blood Pressure                BMI
##               0.609              0.772              0.556              0.609
## Oxygen Saturation        Cholesterol      Triglycerides              HbA1c
##               0.660              0.652              0.852              0.765
## Physical Activity         Diet Score       Stress Level        Sleep Hours
##               0.595              0.579              0.422              0.496
```

```r
library(tibble)
library(dplyr)

table_m4 <- tibble(
  Variable = rownames(fit1_vectors),
  PC1 = round(L1, 3),
  PC2 = round(L2, 3),
  PC3 = round(L3, 3),
  PC4 = round(L4, 3),
  Communality_m4 = round(comm_m4, 3)
)

table_m4
```

```
## # A tibble: 12 x 6
##    Variable             PC1    PC2    PC3    PC4 Communality_m4
##    <chr>              <dbl>  <dbl>  <dbl>  <dbl>          <dbl>
##  1 Age               -0.38  -0.38   0.109 -0.555          0.609
##  2 Glucose           -0.543  0.535  0.373 -0.228          0.772
##  3 Blood Pressure    -0.439 -0.536  0.045 -0.27           0.555
##  4 BMI               -0.416  0.153 -0.583  0.22           0.585
##  5 Oxygen Saturation  0.035  0.105 -0.442 -0.664          0.648
##  6 Cholesterol       -0.414 -0.453 -0.147  0.054          0.401
##  7 Triglycerides     -0.462 -0.183 -0.289  0.163          0.357
##  8 HbA1c             -0.581  0.549  0.314 -0.166          0.765
##  9 Physical Activity  0.664 -0.147  0.322 -0.147          0.587
## 10 Diet Score         0.727 -0.139  0.122 -0.111          0.575
## 11 Stress Level      -0.345 -0.401  0.371  0.032          0.419
## 12 Sleep Hours        0.386  0.199 -0.389 -0.382          0.486
```

```r
table_m5 <- tibble(
  Variable = rownames(fit1_vectors),
  PC1 = round(L1, 3),
  PC2 = round(L2, 3),
```

```r
  PC3 = round(L3, 3),
  PC4 = round(L4, 3),
  PC5 = round(L5, 3),
  Communality_m5 = round(comm_m5, 3)
)

table_m5
```

```
## # A tibble: 12 x 7
##    Variable             PC1    PC2    PC3    PC4    PC5 Communality_m5
##    <chr>              <dbl>  <dbl>  <dbl>  <dbl>  <dbl>          <dbl>
##  1 Age               -0.38  -0.38   0.109 -0.555 -0.009          0.609
##  2 Glucose           -0.543  0.535  0.373 -0.228 -0.008          0.772
##  3 Blood Pressure    -0.439 -0.536  0.045 -0.27   0.031          0.556
##  4 BMI               -0.416  0.153 -0.583  0.22  -0.156          0.609
##  5 Oxygen Saturation  0.035  0.105 -0.442 -0.664  0.111          0.66
##  6 Cholesterol       -0.414 -0.453 -0.147  0.054 -0.501          0.652
##  7 Triglycerides     -0.462 -0.183 -0.289  0.163  0.703          0.852
##  8 HbA1c             -0.581  0.549  0.314 -0.166 -0.012          0.765
##  9 Physical Activity  0.664 -0.147  0.322 -0.147  0.087          0.595
## 10 Diet Score         0.727 -0.139  0.122 -0.111  0.063          0.579
## 11 Stress Level      -0.345 -0.401  0.371  0.032  0.05           0.422
## 12 Sleep Hours        0.386  0.199 -0.389 -0.382 -0.101          0.496
```

Proportion of variance

```r
prop_m4 <- (sum(L1^2) + sum(L2^2) + sum(L3^2) + sum(L4^2)) / length(L1)
prop_m4
```

```
## [1] 0.5632188
```

```r
prop_m5 <- (sum(L1^2) + sum(L2^2) + sum(L3^2) + sum(L4^2) + sum(L5^2)) / length(L1)
prop_m5
```

```
## [1] 0.6305853
```

Varimax rotation

```r
#m = 4
loadings_m4 <- cbind(L1, L2, L3, L4)
rot4 <- varimax(loadings_m4, normalize = FALSE)
rot4
```

```
## $loadings
##
## Loadings:
##                   L1     L2     L3     L4
## Age                      0.159  0.741 -0.184
## Glucose                  0.876
## Blood Pressure     -0.133         0.731
## BMI               -0.748        -0.107 -0.118
## Oxygen Saturation                0.163 -0.786
## Cholesterol       -0.363 -0.147  0.472  0.157
## Triglycerides     -0.542         0.228  0.100
```

```
## HbA1c             -0.138  0.863
## Physical Activity  0.707 -0.279
## Diet Score         0.601 -0.407 -0.177 -0.127
## Stress Level                     0.494  0.408
## Sleep Hours        0.111 -0.156 -0.202 -0.639
##
##                   L1    L2    L3    L4
## SS loadings    1.903 1.836 1.723 1.296
## Proportion Var 0.159 0.153 0.144 0.108
## Cumulative Var 0.159 0.312 0.455 0.563
##
## $rotmat
##             [,1]       [,2]       [,3]       [,4]
## [1,]  0.65029969 -0.5522586 -0.4801441 -0.2039175
## [2,] -0.05355636  0.6509681 -0.7130286 -0.2548774
## [3,]  0.67348556  0.4481643  0.1559712  0.5667795
## [4,] -0.34736037 -0.2653295 -0.4865438  0.7564497
```

```
#m = 5
loadings_m5 <- cbind(L1, L2, L3, L4, L5)
rot5 <- varimax(loadings_m5, normalize = FALSE)
rot5
```

```
## $loadings
##
## Loadings:
##                   L1     L2     L3     L4     L5
## Age               -0.160         0.734 -0.194
## Glucose           -0.876
## Blood Pressure                   0.711         0.206
## BMI                      -0.743 -0.117 -0.118  0.170
## Oxygen Saturation                0.139 -0.797
## Cholesterol        0.148 -0.535  0.526  0.187 -0.182
## Triglycerides            -0.159                0.904
## HbA1c             -0.866 -0.116
## Physical Activity  0.298  0.664               -0.246
## Diet Score         0.425  0.552 -0.157 -0.116 -0.236
## Stress Level                     0.483  0.393  0.139
## Sleep Hours        0.166        -0.180 -0.621 -0.222
##
##                   L1    L2    L3    L4    L5
## SS loadings    1.872 1.640 1.660 1.277 1.117
## Proportion Var 0.156 0.137 0.138 0.106 0.093
## Cumulative Var 0.156 0.293 0.431 0.537 0.631
##
## $rotmat
##            [,1]        [,2]       [,3]        [,4]       [,5]
## [1,]  0.5747153  0.55544096 -0.4446157 -0.17987728 -0.3621446
## [2,] -0.6473689 -0.06431681 -0.7059901 -0.24332019 -0.1383840
## [3,] -0.4308440  0.62378420  0.1876149  0.57529984 -0.2430997
## [4,]  0.2530829 -0.30946473 -0.4953197  0.75439321  0.1604054
## [5,] -0.0306314  0.44997060 -0.1528199 -0.09143538  0.8745706
```

Running Maximum Likelihood Factor Analysis

Running analysis

```r
#m = 4
fa_ml_4 <- factanal(healthcare_risk_cont, factors = 4, rotation = "varimax", scores = "regression")

#m = 5
fa_ml_5 <- factanal(healthcare_risk_cont, factors = 5, rotation = "varimax", scores = "regression")
```

Fit summaries

```r
load_m4 <- as.data.frame(fa_ml_4$loadings[,1:4])
colnames(load_m4) <- paste0("F", 1:4)

table_ml4 <- tibble(
  Variable = rownames(load_m4),
  F1_ML = round(load_m4$F1, 3),
  F2_ML = round(load_m4$F2, 3),
  F3_ML = round(load_m4$F3, 3),
  F4_ML = round(load_m4$F4, 3),
  Uniqueness = round(fa_ml_4$uniquenesses, 3)
)

table_ml4
```

```
## # A tibble: 12 x 6
##    Variable           F1_ML  F2_ML  F3_ML  F4_ML Uniqueness
##    <chr>              <dbl>  <dbl>  <dbl>  <dbl>      <dbl>
##  1 Age                0.107  0.04   0.552  0.09       0.674
##  2 Glucose            0.794  0.102  0.061 -0.04       0.355
##  3 Blood Pressure     0.004  0.123  0.603 -0.062      0.617
##  4 BMI                0.032  0.535 -0.02   0.084      0.705
##  5 Oxygen Saturation  0.013  0.041  0.061  0.452      0.79
##  6 Cholesterol       -0.058  0.258  0.353 -0.121      0.791
##  7 Triglycerides      0.019  0.359  0.211 -0.078      0.821
##  8 HbA1c              0.759  0.192  0.029 -0.06       0.383
##  9 Physical Activity -0.198 -0.661 -0.11   0.035      0.51
## 10 Diet Score        -0.301 -0.573 -0.188  0.136      0.527
## 11 Stress Level       0.058  0.038  0.344 -0.279      0.799
## 12 Sleep Hours       -0.105 -0.123 -0.188  0.433      0.751
```

```r
load_m5 <- as.data.frame(fa_ml_5$loadings[,1:5])
colnames(load_m5) <- paste0("F", 1:5)

table_ml5 <- tibble(
  Variable = rownames(load_m5),
  F1_ML = round(load_m5$F1, 3),
  F2_ML = round(load_m5$F2, 3),
  F3_ML = round(load_m5$F3, 3),
  F4_ML = round(load_m5$F4, 3),
  F5_ML = round(load_m5$F5, 3),
  Uniqueness = round(fa_ml_5$uniquenesses, 3)
)
```

```r
table_ml5
```

```
## # A tibble: 12 x 7
##    Variable           F1_ML  F2_ML  F3_ML  F4_ML  F5_ML Uniqueness
##    <chr>              <dbl>  <dbl>  <dbl>  <dbl>  <dbl>      <dbl>
##  1 Age                0.108  0.027  0.578  0.1   -0.252      0.58
##  2 Glucose            0.809  0.098  0.053 -0.04   0.004      0.332
##  3 Blood Pressure     0.011  0.137  0.637 -0.061  0.141      0.552
##  4 BMI                0.037  0.545 -0.019  0.099  0.089      0.683
##  5 Oxygen Saturation  0.012  0.041  0.053  0.429 -0.03       0.81
##  6 Cholesterol       -0.054  0.264  0.338 -0.128 -0.002      0.796
##  7 Triglycerides      0.023  0.363  0.204 -0.08   0.034      0.818
##  8 HbA1c              0.747  0.192  0.02  -0.063 -0.029      0.4
##  9 Physical Activity -0.198 -0.663 -0.093  0.041  0.122      0.496
## 10 Diet Score        -0.302 -0.57  -0.172  0.142  0.084      0.527
## 11 Stress Level       0.06   0.046  0.328 -0.287 -0.01       0.804
## 12 Sleep Hours       -0.105 -0.124 -0.178  0.444  0.036      0.743
```

Comparing Two Methods

m = 4 model

```r
comparison_m4 <- table_m4 %>%
  select(Variable, PC1, PC2, PC3, PC4) %>%
  left_join(
    table_ml4 %>% select(Variable, F1_ML, F2_ML, F3_ML, F4_ML, Uniqueness),
    by = "Variable"
  )

comparison_m4
```

```
## # A tibble: 12 x 10
##    Variable      PC1    PC2    PC3    PC4  F1_ML  F2_ML  F3_ML  F4_ML Uniqueness
##    <chr>       <dbl>  <dbl>  <dbl>  <dbl>  <dbl>  <dbl>  <dbl>  <dbl>      <dbl>
##  1 Age        -0.38  -0.38   0.109 -0.555  0.107  0.04   0.552  0.09       0.674
##  2 Glucose    -0.543  0.535  0.373 -0.228  0.794  0.102  0.061 -0.04       0.355
##  3 Blood Pre~ -0.439 -0.536  0.045 -0.27   0.004  0.123  0.603 -0.062      0.617
##  4 BMI        -0.416  0.153 -0.583  0.22   0.032  0.535 -0.02   0.084      0.705
##  5 Oxygen Sa~  0.035  0.105 -0.442 -0.664  0.013  0.041  0.061  0.452      0.79
##  6 Cholester~ -0.414 -0.453 -0.147  0.054 -0.058  0.258  0.353 -0.121      0.791
##  7 Triglycer~ -0.462 -0.183 -0.289  0.163  0.019  0.359  0.211 -0.078      0.821
##  8 HbA1c      -0.581  0.549  0.314 -0.166  0.759  0.192  0.029 -0.06       0.383
##  9 Physical ~  0.664 -0.147  0.322 -0.147 -0.198 -0.661 -0.11   0.035      0.51
## 10 Diet Score  0.727 -0.139  0.122 -0.111 -0.301 -0.573 -0.188  0.136      0.527
## 11 Stress Le~ -0.345 -0.401  0.371  0.032  0.058  0.038  0.344 -0.279      0.799
## 12 Sleep Hou~  0.386  0.199 -0.389 -0.382 -0.105 -0.123 -0.188  0.433      0.751
```

m = 5 model

```r
comparison_m5 <- table_m5 %>%
  select(Variable, PC1, PC2, PC3, PC4, PC5) %>%
  left_join(
```

```r
    table_ml5 %>% select(Variable, F1_ML, F2_ML, F3_ML, F4_ML, F5_ML, Uniqueness),
    by = "Variable"
  )

comparison_m5
```

```
## # A tibble: 12 x 12
##    Variable          PC1    PC2    PC3    PC4    PC5  F1_ML  F2_ML  F3_ML  F4_ML
##    <chr>           <dbl>  <dbl>  <dbl>  <dbl>  <dbl>  <dbl>  <dbl>  <dbl>  <dbl>
##  1 Age            -0.38  -0.38   0.109 -0.555 -0.009  0.108  0.027  0.578  0.1
##  2 Glucose        -0.543  0.535  0.373 -0.228 -0.008  0.809  0.098  0.053 -0.04
##  3 Blood Pressure -0.439 -0.536  0.045 -0.27   0.031  0.011  0.137  0.637 -0.061
##  4 BMI            -0.416  0.153 -0.583  0.22  -0.156  0.037  0.545 -0.019  0.099
##  5 Oxygen Satura~  0.035  0.105 -0.442 -0.664  0.111  0.012  0.041  0.053  0.429
##  6 Cholesterol    -0.414 -0.453 -0.147  0.054 -0.501 -0.054  0.264  0.338 -0.128
##  7 Triglycerides  -0.462 -0.183 -0.289  0.163  0.703  0.023  0.363  0.204 -0.08
##  8 HbA1c          -0.581  0.549  0.314 -0.166 -0.012  0.747  0.192  0.02  -0.063
##  9 Physical Acti~  0.664 -0.147  0.322 -0.147  0.087 -0.198 -0.663 -0.093  0.041
## 10 Diet Score      0.727 -0.139  0.122 -0.111  0.063 -0.302 -0.57  -0.172  0.142
## 11 Stress Level   -0.345 -0.401  0.371  0.032  0.05   0.06   0.046  0.328 -0.287
## 12 Sleep Hours     0.386  0.199 -0.389 -0.382 -0.101 -0.105 -0.124 -0.178  0.444
## # i 2 more variables: F5_ML <dbl>, Uniqueness <dbl>
```

Comparing Scores

```r
scores_pca <- as.data.frame(Xs %*% fit1_vectors[, 1:4])
colnames(scores_pca) <- paste0("PC", 1:4)

# ML factor scores (already from factanal with m = 4)
scores_ml <- as.data.frame(fa_ml_4$scores)
colnames(scores_ml) <- paste0("F", 1:4)

scores_compare<- bind_cols(scores_pca, scores_ml)

options(scipen = 999)    # turn off scientific notation

# rounding EVERYTHING to 3-4 decimals (you choose)
scores_pca_round <- round(scores_pca, 3)
scores_ml_round <- round(scores_ml, 3)

scores_compare_round <- bind_cols(scores_pca_round, scores_ml_round)

# Correlations between PCA and MLFA scores
cor(scores_compare_round)
```

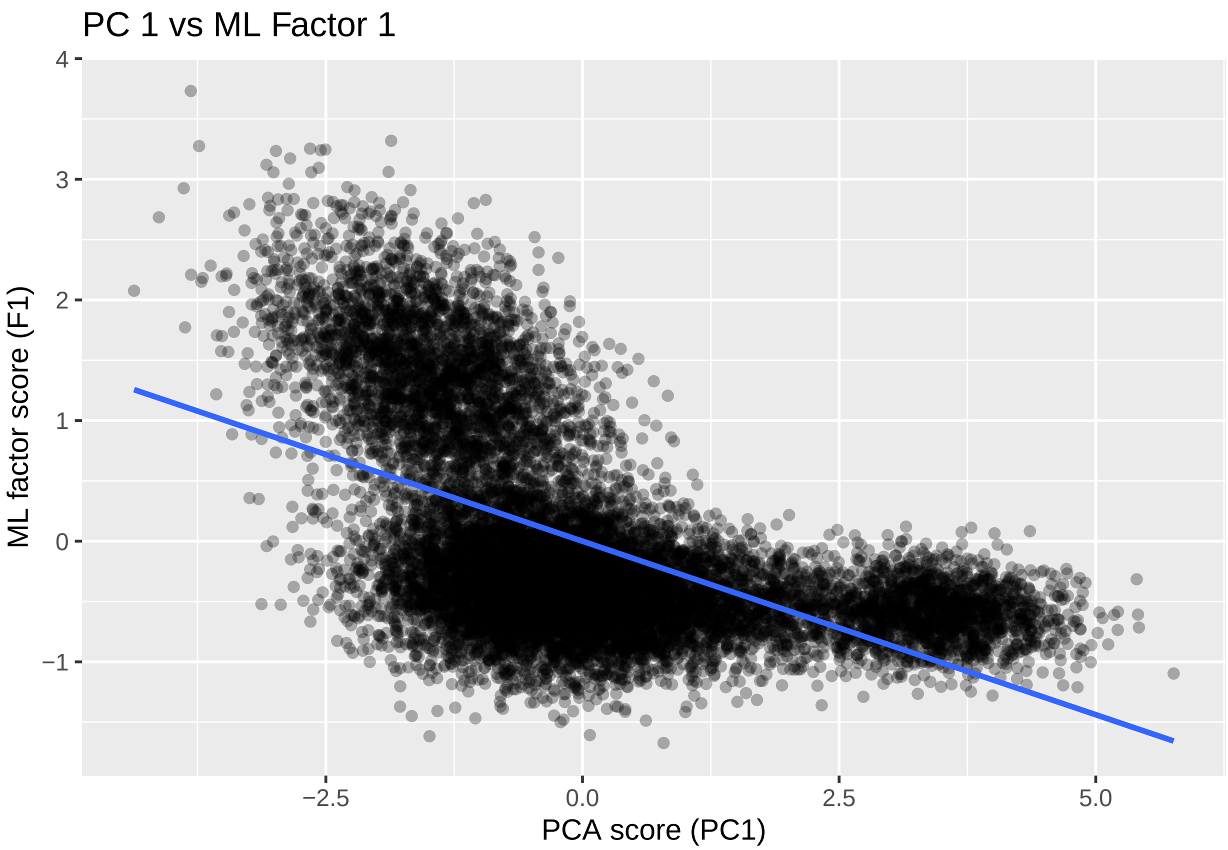
```
##                 PC1             PC2             PC3             PC4          F1
## PC1  1.000000000000 -0.000001500745  0.000002581956  0.000002799952 -0.55073381
## PC2 -0.000001500745  1.000000000000  0.000002068458 -0.000000578048  0.64125101
## PC3  0.000002581956  0.000002068458  1.000000000000  0.000004065305  0.43113933
## PC4  0.000002799952 -0.000000578048  0.000004065305  1.000000000000 -0.26194308
## F1  -0.550733807808  0.641251006537  0.431139327335 -0.261943082435  1.00000000
## F2  -0.743354338334  0.096665404303 -0.561299523050  0.281291212036  0.12285304
## F3  -0.554306309375 -0.701315167020  0.136025169431 -0.409358085051  0.02259041
## F4   0.289917825217  0.258672272951 -0.588314298878 -0.702964453014 -0.05801999
```

```
##              F2          F3          F4
## PC1 -0.74335434 -0.55430631  0.28991783
## PC2  0.09666540 -0.70131517  0.25867227
## PC3 -0.56129952  0.13602517 -0.58831430
## PC4  0.28129121 -0.40935809 -0.70296445
## F1   0.12285304  0.02259041 -0.05801999
## F2   1.00000000  0.14739594 -0.06240772
## F3   0.14739594  1.00000000 -0.13101015
## F4  -0.06240772 -0.13101015  1.00000000
```

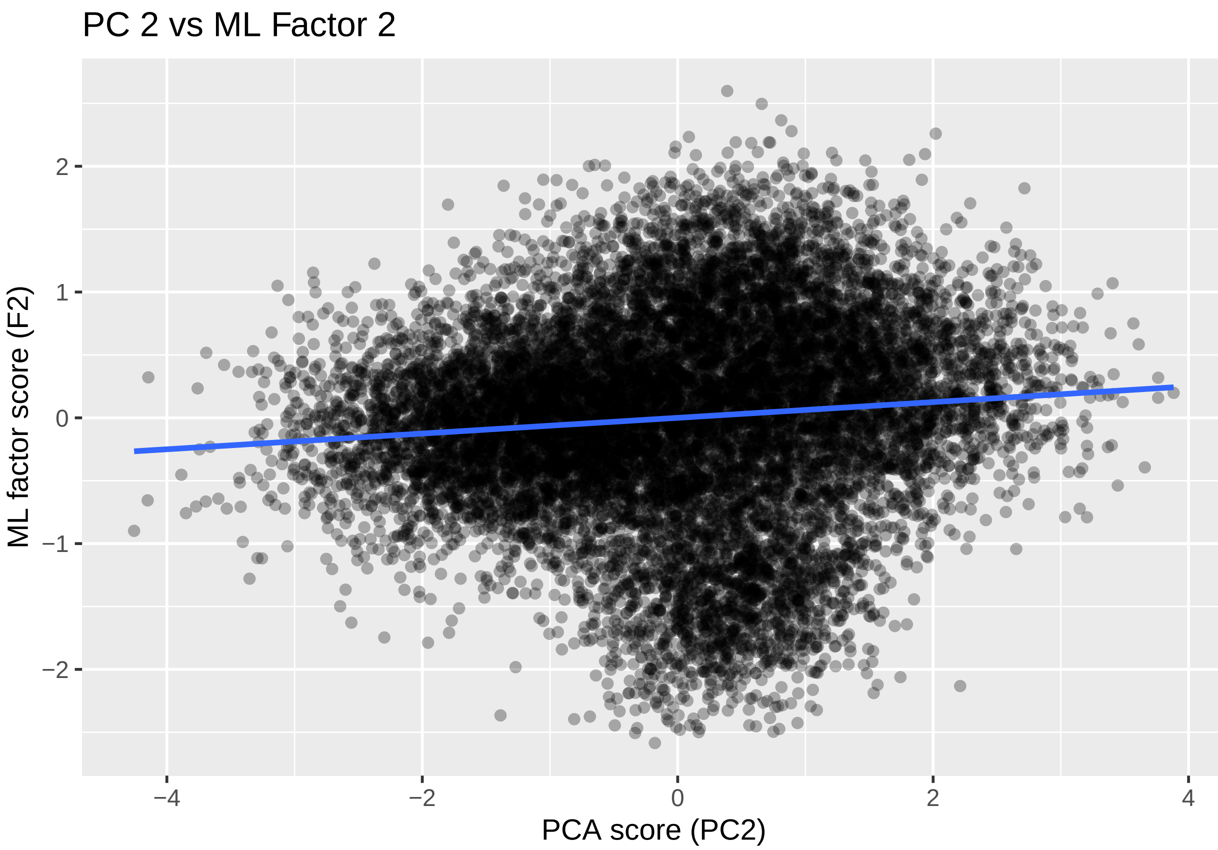
```r
library(ggplot2)

for (i in 1:4) {
  p <- ggplot(scores_compare,
              aes_string(x = paste0("PC", i), y = paste0("F", i))) +
    geom_point(alpha = 0.3) +
    geom_smooth(method = "lm", se = FALSE) +
    labs(
      title = paste("PC", i, "vs ML Factor", i),
      x = paste("PCA score (PC", i, ")", sep = ""),
      y = paste("ML factor score (F", i, ")", sep = "")
    )
  print(p)
}
```

```
## Warning: `aes_string()` was deprecated in ggplot2 3.0.0.
## i Please use tidy evaluation idioms with `aes()`.
## i See also `vignette("ggplot2-in-packages")` for more information.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```

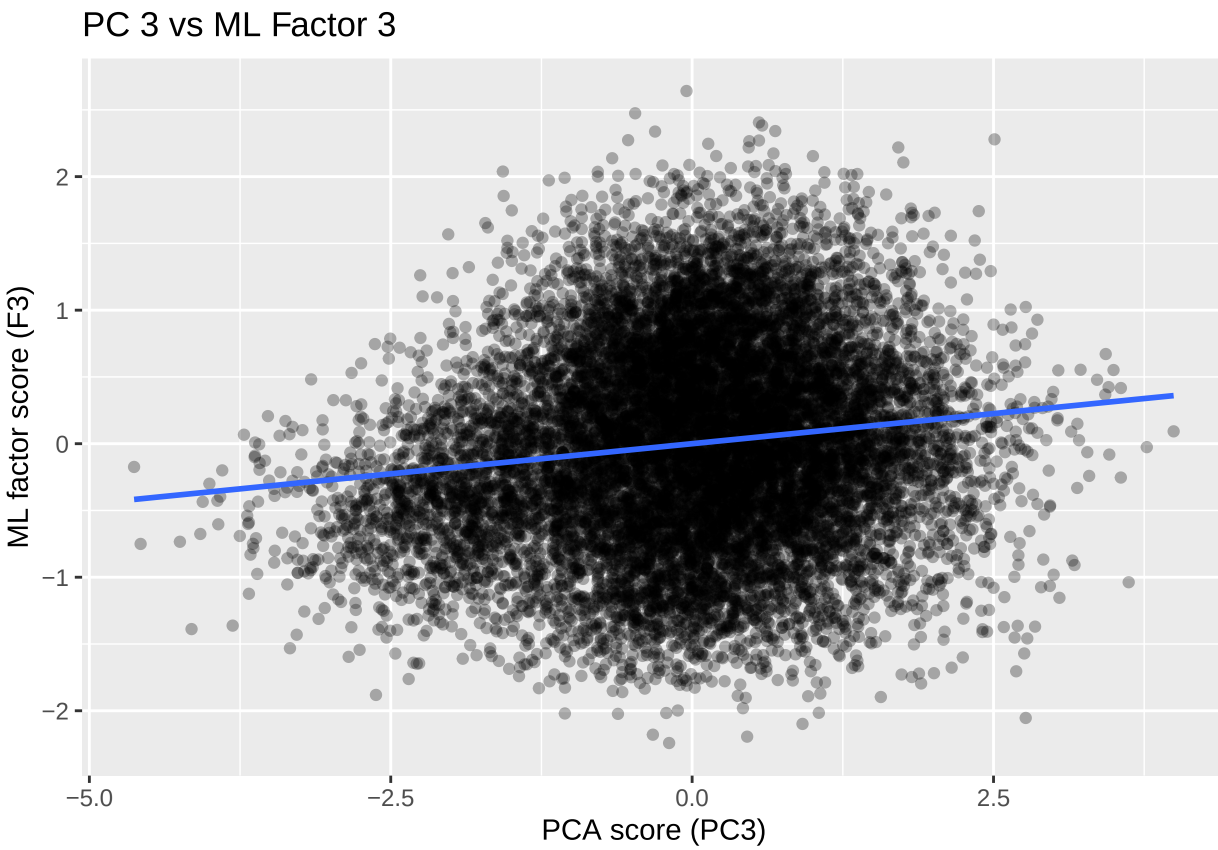
```
## `geom_smooth()` using formula = 'y ~ x'
```




```
## `geom_smooth()` using formula = 'y ~ x'
```

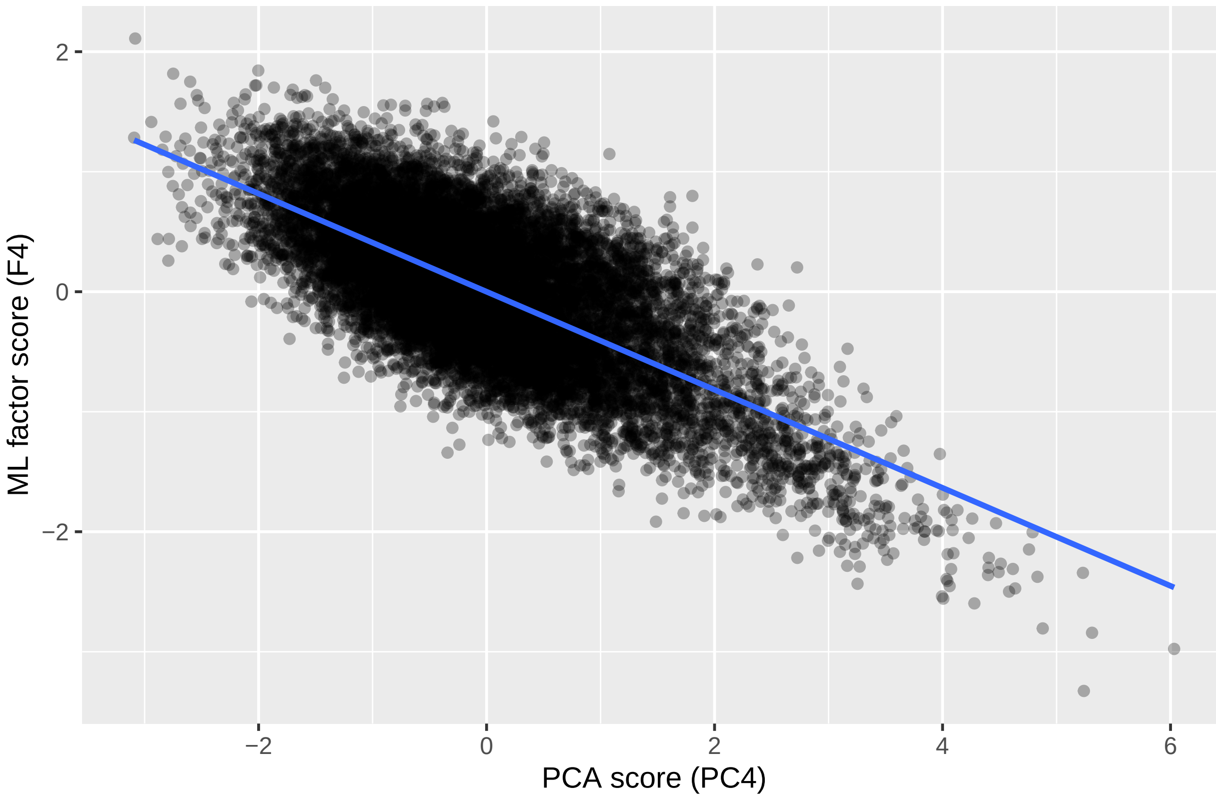



```
## `geom_smooth()` using formula = 'y ~ x'
```




```
## `geom_smooth()` using formula = 'y ~ x'
```

PC 4 vs ML Factor 4



To evaluate how closely the two extraction methods recover similar latent structure, we compared the PCA scores (PC1–PC4) obtained via eigen-decomposition with the MLFA regression scores (F1–F4). The correlation matrix shows several clear patterns. First, PC1 is strongly negatively correlated with both F1 (r = –0.551) and F3 (r = –0.554), and moderately with F2 (r = –0.744), indicating that PCA's first component blends variance coming from multiple underlying latent constructs. In contrast, MLFA separates these influences across factors, which aligns with the more interpretable loading patterns observed previously.

Second, PC2 correlates positively with F1 (r = 0.641) and negatively with F3 (r = –0.701), suggesting that PCA's second component captures a mixture of glycemic-related variance (F1) and age/BP-related variance (F3), but again does not isolate these domains as cleanly as the MLFA solution. PC3 shows modest correlations with F1 (r = 0.431) and F4 (r = –0.588), while PC4 is moderately correlated with F2 (r = 0.281) and strongly negatively correlated with F4 (r = –0.703). Altogether, these patterns indicate that while PCA components reflect broad variance gradients, they do not map cleanly onto the interpretable physiological constructs that emerge under MLFA.

In summary, the factor score comparison demonstrates that MLFA provides a more distinct and psychologically interpretable separation of latent health domains, whereas PCA components are mixtures of multiple ML factors. This supports the decision to use MLFA factors in the downstream regression analyses and for conceptual interpretation of latent health profiles.

Given goal of this project, going to move forward with MLE structure.

Running regression with factors

```
library(nnet)
library(broom)
```

```r
healthcare_risk_fa_reg <- healthcare_risk_clean %>%
  select(Age, Glucose, `Blood Pressure`, BMI, `Oxygen Saturation`,
    Cholesterol, Triglycerides, HbA1c,
    `Physical Activity`, `Diet Score`,
    `Stress Level`, `Sleep Hours`, `Medical Condition`)

scores_ml <- as.data.frame(fa_ml_4$scores) #this was already run on only continuous variables
colnames(scores_ml) <- paste0("F", 1:4)

final_reg_data <- bind_cols(healthcare_risk_fa_reg, scores_ml)

#setting "healthy" as reference
final_reg_data <- final_reg_data %>%
  mutate(MedicalCondition = relevel(
    factor(`Medical Condition`),
    ref = "Healthy"
  ))

# run regression
reg_fit <- multinom(
  MedicalCondition ~ F1 + F2 + F3 + F4,
  data = final_reg_data
)
```

```
## # weights:  42 (30 variable)
## initial  value 25987.630041
## iter  10 value 6825.667653
## iter  20 value 6256.328644
## iter  30 value 5685.715085
## iter  40 value 5521.905528
## iter  50 value 5508.802172
## final  value 5508.785389
## converged
```

Regression Output

```r
reg_results <- tidy(reg_fit, exponentiate = TRUE, conf.int = TRUE)
reg_results
```

```
## # A tibble: 30 x 8
##    y.level    term        estimate std.error statistic  p.value conf.low conf.high
##    <chr>      <chr>          <dbl>     <dbl>     <dbl>    <dbl>    <dbl>     <dbl>
##  1 Arthritis (Intercep~  1.46e+5     0.706     16.9  9.30e-64 3.67e+4   5.84e+5
##  2 Arthritis F1          2.07e+2     0.565      9.44 3.59e-21 6.84e+1   6.25e+2
##  3 Arthritis F2          5.59e+3     0.448     19.3  1.10e-82 2.32e+3   1.34e+4
##  4 Arthritis F3          1.21e+3     0.374     19.0  1.79e-80 5.81e+2   2.51e+3
##  5 Arthritis F4          8.10e-2     0.392     -6.40 1.51e-10 3.75e-2   1.75e-1
##  6 Asthma    (Intercep~  4.66e+3     0.688     12.3  1.29e-34 1.21e+3   1.80e+4
##  7 Asthma    F1          2.54e+1     0.529      6.11 9.66e-10 9.02e+0   7.18e+1
##  8 Asthma    F2          1.83e+2     0.407     12.8  1.76e-37 8.24e+1   4.07e+2
##  9 Asthma    F3          7.45e+0     0.313      6.41 1.44e-10 4.03e+0   1.38e+1
## 10 Asthma    F4          1.40e-3     0.388    -16.9  1.95e-64 6.53e-4   2.99e-3
## # i 20 more rows
```

```r
#scaling factors since regression output is not realistic
final_reg_data_scale <- final_reg_data %>%
  mutate(
    F1 = scale(F1),
    F2 = scale(F2),
    F3 = scale(F3),
    F4 = scale(F4)
  )

reg_fit_scale <- multinom(
  MedicalCondition ~ F1 + F2 + F3 + F4,
  data = final_reg_data_scale
)
```

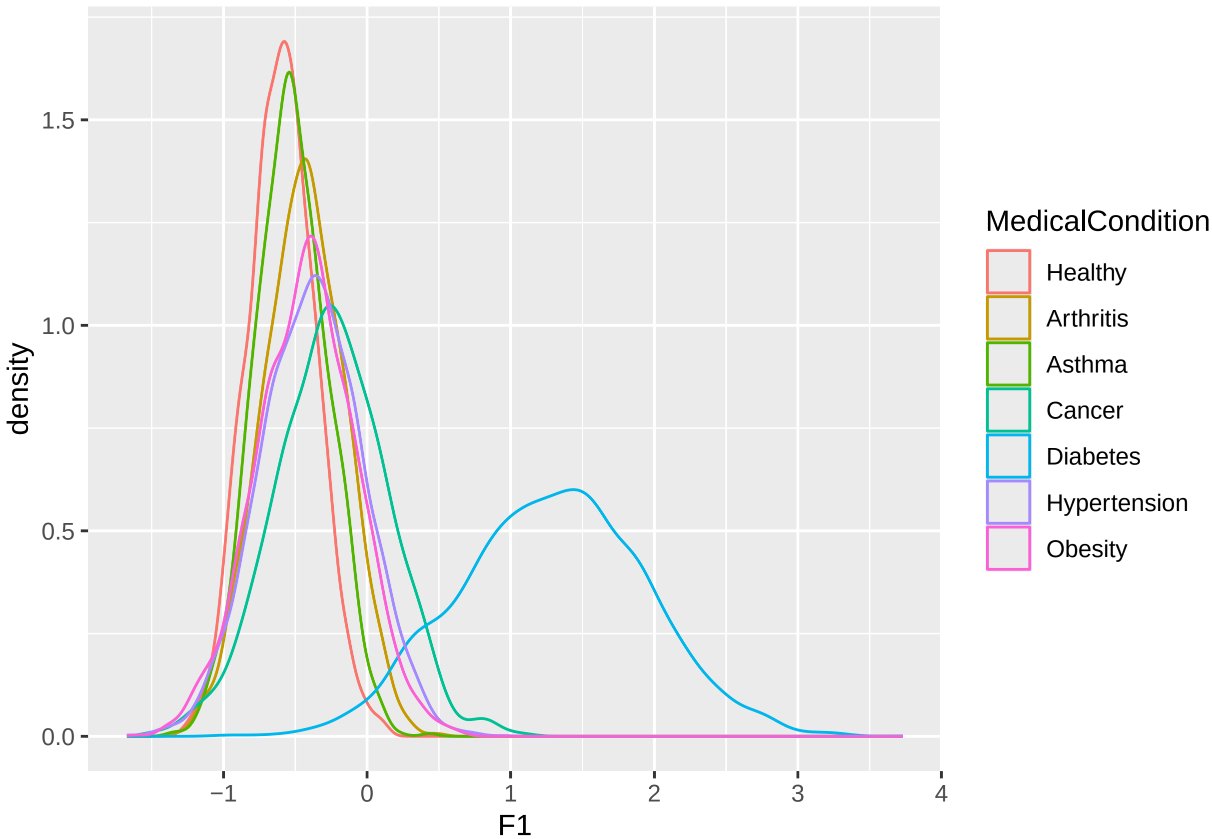
```
## # weights:  42 (30 variable)
## initial  value 25987.630041
## iter  10 value 7029.198959
## iter  20 value 6113.747890
## iter  30 value 5700.535169
## iter  40 value 5516.057806
## iter  50 value 5508.785677
## final  value 5508.785379
## converged
```

```r
reg_results_scale <- tidy(reg_fit_scale, exponentiate = TRUE, conf.int = TRUE)
reg_results_scale
```

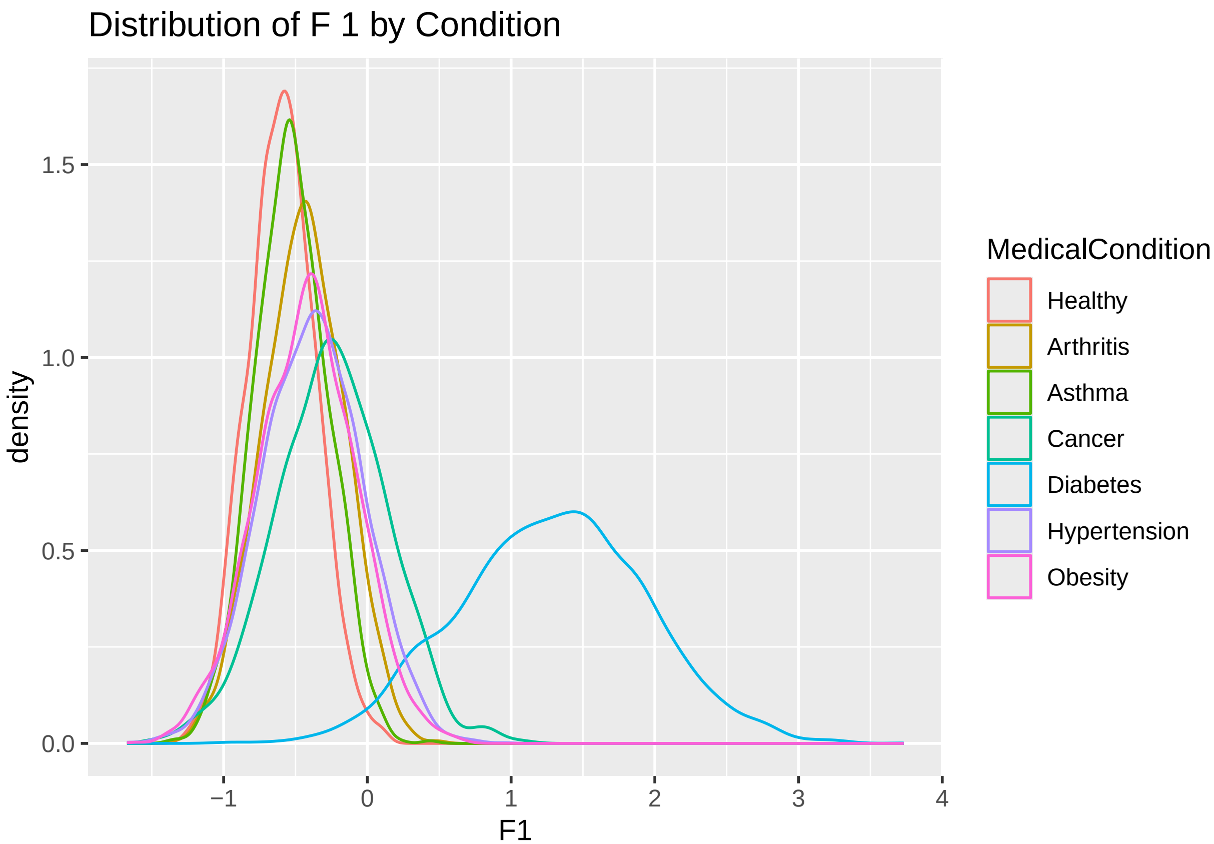
```
## # A tibble: 30 x 8
##    y.level   term       estimate std.error statistic  p.value conf.low conf.high
##    <chr>     <chr>         <dbl>     <dbl>     <dbl>    <dbl>    <dbl>     <dbl>
##  1 Arthritis (Intercep~  1.47e+5     0.706     16.9  9.35e-64  3.68e+4   5.84e+5
##  2 Arthritis F1          1.03e+2     0.490      9.44 3.59e-21  3.92e+1   2.68e+2
##  3 Arthritis F2          1.00e+3     0.359     19.3  1.12e-82  4.97e+2   2.03e+3
##  4 Arthritis F3          2.14e+2     0.282     19.0  1.79e-80  1.23e+2   3.72e+2
##  5 Arthritis F4          2.07e-1     0.246     -6.40 1.51e-10  1.28e-1   3.36e-1
##  6 Asthma    (Intercep~  4.66e+3     0.688     12.3  1.30e-34  1.21e+3   1.80e+4
##  7 Asthma    F1          1.66e+1     0.460      6.11 9.69e-10  6.75e+0   4.09e+1
##  8 Asthma    F2          6.49e+1     0.326     12.8  1.77e-37  3.43e+1   1.23e+2
##  9 Asthma    F3          4.56e+0     0.237      6.41 1.44e-10  2.87e+0   7.26e+0
## 10 Asthma    F4          1.63e-2     0.243    -16.9  1.96e-64  1.01e-2   2.63e-2
## # i 20 more rows
```

Still have huge RRRs, checking class separation:

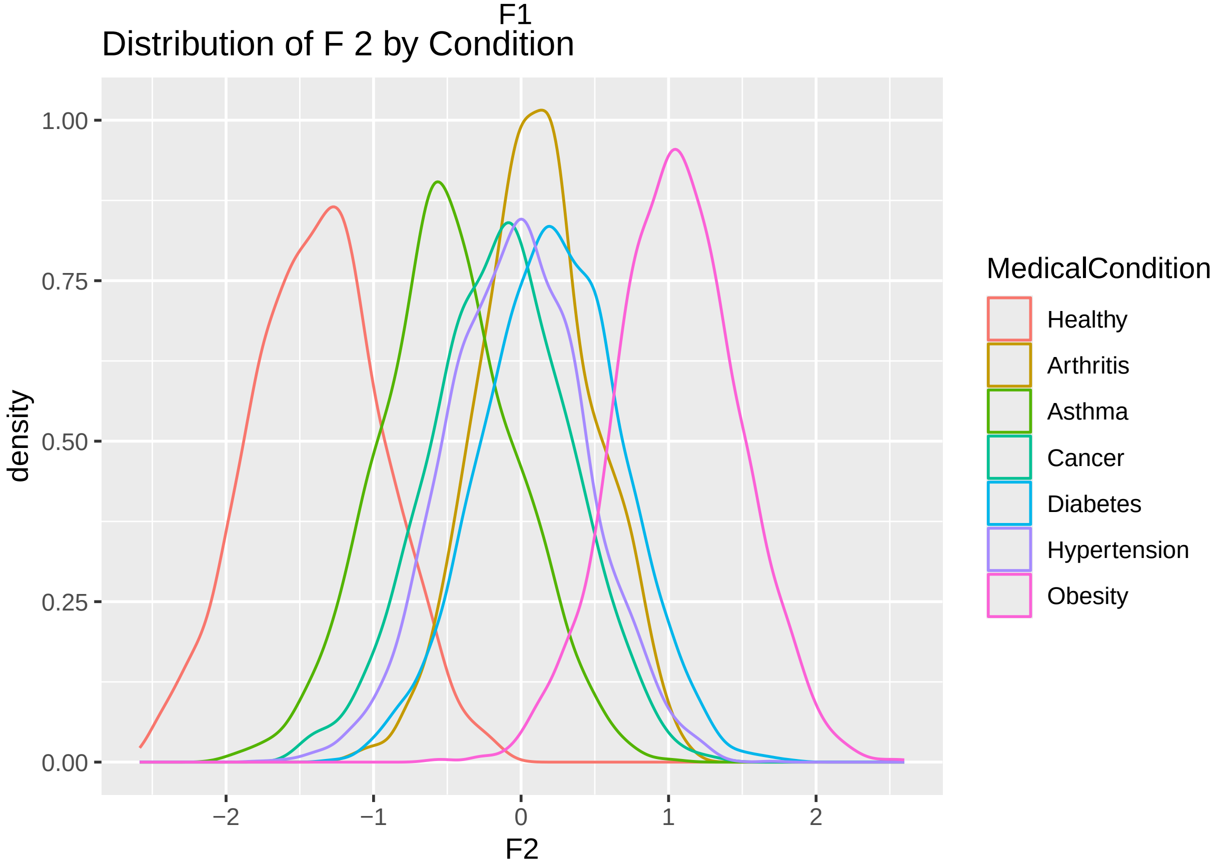
```r
library(ggplot2)

ggplot(final_reg_data, aes(x = F1, color = MedicalCondition)) +
  geom_density()
```




```r
for (i in 1:4) {
  print(
    ggplot(final_reg_data, aes_string(x = paste0("F", i), color = "MedicalCondition")) +
      geom_density() +
      ggtitle(paste("Distribution of F", i, "by Condition"))
  )
}
```

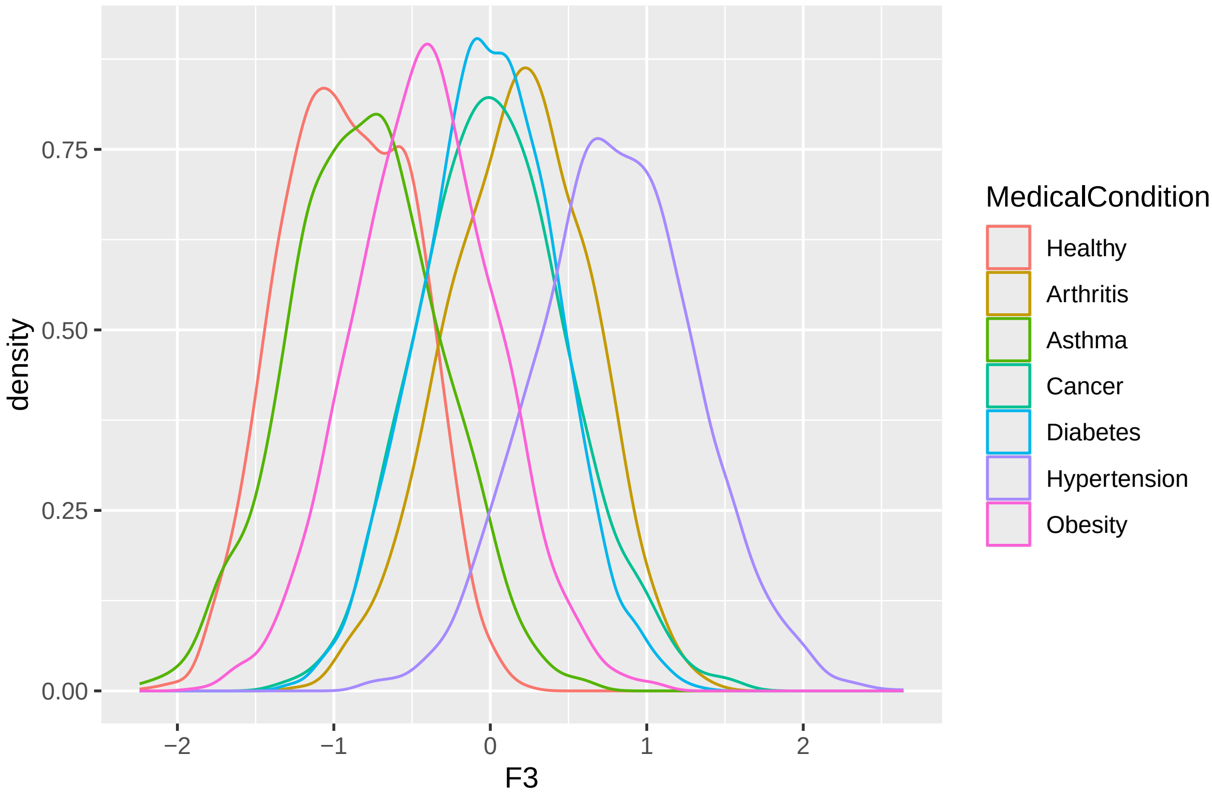


Distribution of F 1 by Condition
density
F1
MedicalCondition
Healthy
Arthritis
Asthma
Cancer
Diabetes
Hypertension
Obesity

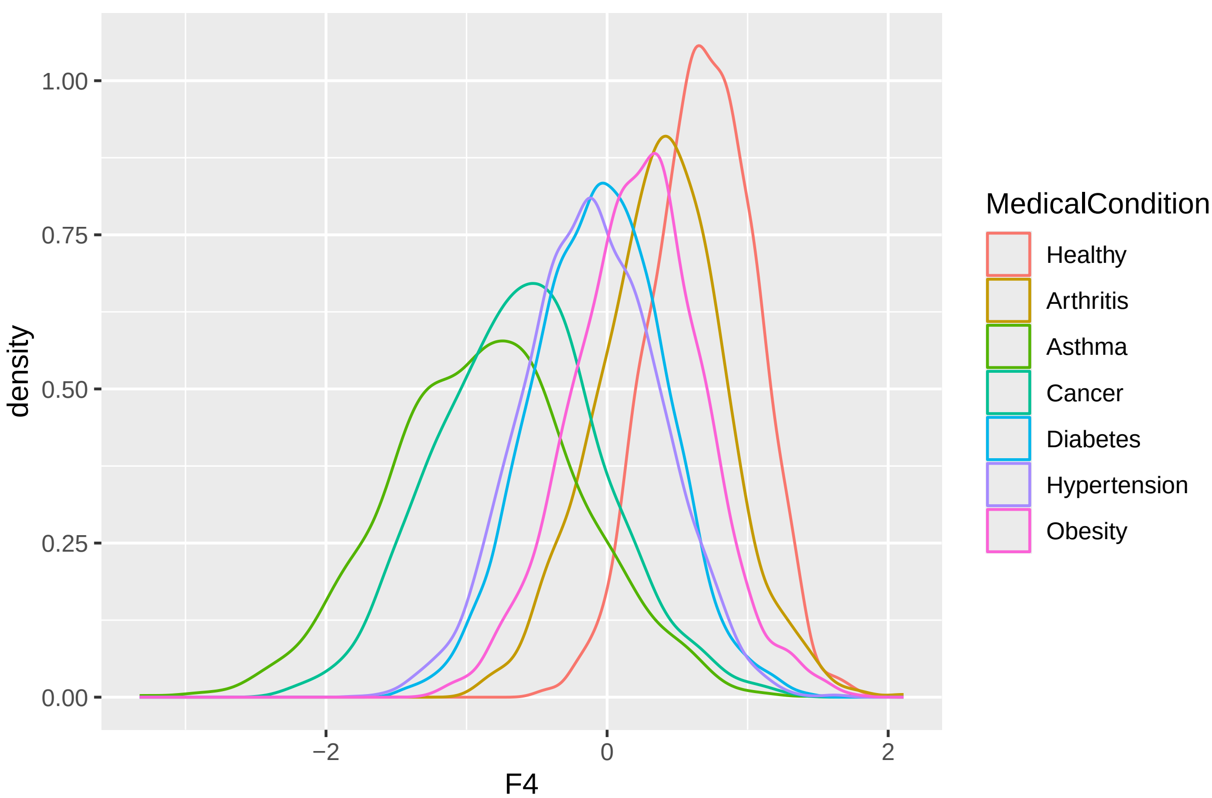


Distribution of F 2 by Condition
density
F2
MedicalCondition
Healthy
Arthritis
Asthma
Cancer
Diabetes
Hypertension
Obesity

Distribution of F 3 by Condition



Distribution of F 4 by Condition



Running regularized multinomial logistic regression, which can handle separation

```r
library(glmnet)
```

```
## Loading required package: Matrix

##
## Attaching package: 'Matrix'

## The following objects are masked from 'package:tidyr':
##
##     expand, pack, unpack

## Loaded glmnet 4.1-10
```

```r
X <- as.matrix(final_reg_data[, c("F1","F2","F3","F4")])
y <- final_reg_data$MedicalCondition

fit_reg_glm <- cv.glmnet(
  X, y,
  family = "multinomial",
  alpha = 1
)

fit_reg_glm$lambda.min
```

```
## [1] 0.00008704828
```

```r
coef(fit_reg_glm, s = "lambda.min")
```

```
## $Healthy
## 5 x 1 sparse Matrix of class "dgCMatrix"
##             lambda.min
## (Intercept)  -8.786619
## F1           -5.731754
## F2           -7.865048
## F3           -5.908198
## F4            4.224872
##
## $Arthritis
## 5 x 1 sparse Matrix of class "dgCMatrix"
##             lambda.min
## (Intercept)  2.2310429
## F1          -0.9285252
## F2           0.2832953
## F3           0.8012139
## F4           1.9592042
##
## $Asthma
## 5 x 1 sparse Matrix of class "dgCMatrix"
##             lambda.min
## (Intercept)  -1.052866
## F1           -2.858672
## F2           -2.979729
## F3           -4.116405
```

```
## F4           -2.073230
##
## $Cancer
## 5 x 1 sparse Matrix of class "dgCMatrix"
##              lambda.min
## (Intercept)  2.0906190
## F1           0.6460717
## F2          -0.6998260
## F3           .
## F4          -2.0788695
##
## $Diabetes
## 5 x 1 sparse Matrix of class "dgCMatrix"
##               lambda.min
## (Intercept)  1.90044800
## F1           7.31868431
## F2           2.52346086
## F3           0.65442444
## F4          -0.02120368
##
## $Hypertension
## 5 x 1 sparse Matrix of class "dgCMatrix"
##               lambda.min
## (Intercept) 3.120417433
## F1          0.005022958
## F2          .
## F3          3.291569366
## F4          .
##
## $Obesity
## 5 x 1 sparse Matrix of class "dgCMatrix"
##              lambda.min
## (Intercept)  0.4969577
## F1           .
## F2           5.2421973
## F3          -1.4639606
## F4           0.8466236
```

Interpretation

Regularized multinomial logistic regression was used to estimate the relationship between the four ML-derived latent health factors (F1–F4) and medical condition categories. Because glmnet applies L1 regularization (lasso), many coefficients are shrunk toward zero or set exactly to zero (.), yielding a sparse and interpretable structure. Below, coefficients reflect the log relative risk of belonging to a given disease class (vs. the implicit baseline used by glmnet) per one-unit increase in each factor score.

1. Diabetes

Diabetes shows the strongest and clearest pattern.

F1 has a large positive coefficient (7.32), meaning higher values on the glycemic/metabolic factor strongly increase the likelihood of diabetes.

Modest additional positive contributions from F2 and F3 indicate that dyslipidemia and age/BP-related physiology also play roles.

F4 is effectively zero, meaning this factor does not help distinguish diabetes.

This aligns with the density plots: F1 nearly separates diabetics from all other groups.

2. Hypertension

Hypertension is primarily associated with F3 (3.29), a factor that appears to capture age, blood pressure, and cardiometabolic load.

F1 and F4 are shrunken close to zero.

F2 is removed entirely, meaning lipids/BMI do not distinguish hypertensives once other factors are controlled.

This is consistent with clinical expectations: latent BP/age physiology is the primary driver of this class.

3. Obesity

Obesity is most strongly associated with F2 (5.24), the factor with heavy loadings on BMI, triglycerides, and diet.

F1 is removed, meaning glycemic regulation does not differentiate obesity from other groups.

F3 has a moderate negative effect, implying that individuals with obesity tend to score lower on the BP/age factor compared with other disease groups.

This shows that F2 represents an adiposity/dyslipidemia construct that uniquely predicts obesity.

4. Arthritis

Arthritis shows a mixed profile:

A moderate positive association with F4 (1.96), possibly reflecting stress/sleep physiology.

F1 is negative, suggesting people with arthritis tend to have lower glycemic factor scores than other disease groups.

Small positive contribution from F3 (0.80), reflecting the age-related component of arthritis.

This pattern indicates that arthritis is a more heterogeneous condition with weaker alignment to classic cardiometabolic factors.

5. Asthma

Asthma has consistently negative coefficients across F1–F4. This indicates individuals with asthma tend to have lower latent cardiometabolic risk profiles than other groups. This makes sense: asthma is largely orthogonal to the physiological domains captured by these factors.

6. Cancer

Cancer shows small to moderate coefficients:

Positive association with F1, negative with F2 and F4, and no contribution from F3 (lasso sets it to zero). This suggests cancer patients' profiles differ subtly but not strongly across these latent physiological domains.

7. Healthy

The "Healthy" class serves as a natural contrast group:

Large negative coefficients across all factors simply indicate that higher cardiometabolic or stress-related scores decrease the likelihood of being in the healthy category.

This reflects the expected pattern: lower factor scores correspond to better overall physiological status.